

Candidate's name:

Signature:COMBINATION:

P525/2
CHEMISTRY
Paper 2
(Practical)
March, 2026
3 hours

THE CHEMISTRY DEPARTMENT
PRE REGISTRATION EXAMINATION

S6 CHEMISTRY

Paper 2 (Practical)

2 hours

INSTRUCTIONS TO CANDIDATES:

*This paper consists of **two compulsory** examination items. Responses to these items are to be written in the spaces provided. Use **blue** or **black** ink.*

*All working **must** be clearly shown.*

Mathematical tables and silent non-programmable scientific calculators may be used.

*You are **not** allowed to use reference books (i.e. text books, booklets on qualitative analysis etc.)*

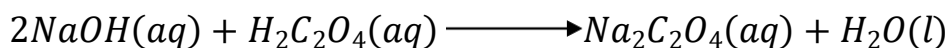
*You are advised to carefully read the items, make sure you have all the apparatus and chemicals you may need and then **plan** accordingly before starting.*

ITEM ONE

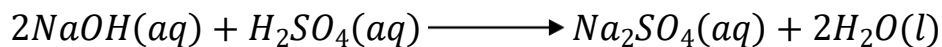
A textile factory uses a copyrighted “*scouring agent*”. Scouring agents remove dirt and impurities from raw cotton and linen fibers, without damaging them in preparation for dyeing. The active ingredient is a precise mixture of oxalic acid dihydrate ($\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$) and sodium oxalate ($\text{Na}_2\text{C}_2\text{O}_4$). The optimal performance requires the sodium oxalate content to be between 20-40% by mass. The workshop manager has become concerned that a recent batch of Mixture *X* is not working as effectively as previous batches. Suspecting an error in its composition, the manager, who is an old girl of your school, has asked your chemistry teacher if students could help verify the composition of the mixture as part of a practical chemistry project.

Your class has been entrusted with this task. You are to determine the percentage by mass of sodium oxalate in the supplied sample of Mixture *X*. *This can be done by reacting an aqueous solution of the mixture of X, made by dissolving exactly 2.40g of the solid in 250cm³ of distilled water, with a fixed volume of sodium hydroxide solution using phenolphthalein indicator. However, a fixed volume of sodium hydroxide solution must first be standardized by dilute sulphuric acid using the same indicator.*

Sodium hydroxide reacts with **only** oxalic acid in the mixture according to the equation;



The base also reacts with sulphuric acid according to the equation;



You are provided with;

- 3.0 g of mixture X.
- 100cm³ of FA1 which is a solution of sodium hydroxide of unknown concentration.
- 100cm³ of FA2 which is a solution containing 0.05moles per litre of dilute sulphuric acid.
- distilled water, sufficient for solution preparation and rinsing
- phenolphthalein indicator
- 250.0 cm³ volumetric flask
- Other laboratory apparatus suitable for a titration experiment. Task;

Task;

As a chemistry learner;

- (a) Design an experiment to standardize *FA1*. (Your design should include; aim, hypothesis, variables, apparatus and materials, procedure, risks and their mitigations, data and data analysis and then conclusion.

Experiment one;

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- (b) Design an experiment to determine the percentage by mass of sodium oxalate in the scouring agent.

Experiment two;

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